

Experimental investigation of high-quality synchronization of coupled oscillators

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(Received 30 November 1999; accepted for publication 2 May 2000)

We describe two experiments in which we investigate the synchronization of coupled periodic oscillators. Each experimental system consists of two identical coupled electronic periodic oscillators that display bursts of desynchronization events similar to those observed previously in coupled chaotic systems. We measure the degree of synchronization as a function of coupling strength. In the first experiment, high-quality synchronization is achieved for all coupling strengths above a critical value. In the second experiment, no high-quality synchronization is observed. We compare our results to the predictions of the several proposed criteria for synchronization. We find that none of the criteria accurately predict the range of coupling strengths over which high-quality synchronization is observed. © 2000 American Institute of Physics. [S1054-1500(00)01203-9]

The once surprising fact that the irregular oscillations of two chaotic oscillators can be synchronized is now well established. However, deterministic chaos is not the only source of irregular oscillations. Recent research shows that stable periodic systems may oscillate irregularly when subject to small random noise [F. Ali and M. Menzinger, *Chaos* 9, 348 (1999); Trefethen *et al.*, *Science* 261, 578 (1993)]. Can a high degree of synchronization be achieved between two systems undergoing irregular oscillations due to small noise rather than deterministic chaos? We address this question here through two experiments on coupled periodic electronic circuits. In each experiment, we observe the degree of synchronization between a pair of coupled oscillators as the coupling strength is increased from zero. In the first experiment, we observe a sudden transition to high-quality synchronization at a critical coupling strength. In the second experiment, no high-quality synchronization is observed over the range of accessible coupling strengths. In addition, we apply to our experimental systems several proposed criteria for high-quality synchronization developed in studies of synchronized chaos. We find that none of these criteria accurately predict the behavior observed in the experiments. These results may provide some guidance in the development of practical applications of synchronized chaos such as secure communication schemes where a criterion for high-quality synchronization is needed.

I. INTRODUCTION

It is now well established that the dynamics of a nonlinear system can become highly irregular when small random

noise is injected into the system. For example, Ali and Menzinger¹ recently showed that a globally stable limit cycle oscillator subject to small amounts of noise can display an explosive divergence of trajectories away from the limit cycle. The origin of this disproportionate response to small perturbations is the fact that the limit cycle is composed of segments of varying local stability, most of which are stable but some of which are highly unstable as shown schematically in Fig. 1. In regions of pronounced local instability, a perturbation may undergo transient growth before decaying asymptotically. A similar behavior is displayed by non-normal linear systems as shown schematically in Fig. 2(a).² Non-normal systems are characterized by nonorthogonal eigenvectors. A small perturbation to such a system expressed as a linear superposition of such vectors may have large coefficients but a small norm due to cancellation as depicted in Fig. 2(b). As shown in Fig. 2(c), when the system evolves in time, the coefficients of the superposition may decay at different exponential rates so the cancellation is lost, causing the norm to increase even though the individual eigenvector components are decaying asymptotically. Again, the result is a transient amplification of a perturbation by a stable dynamical system. These two examples do not exhaust the possible scenarios in which highly irregular oscillations are generated by large noise amplification in a dynamical system. An interesting question is whether the irregular oscillations occurring in two identical noise-amplifying dynamical systems can be synchronized.

The primary objective of this paper is to present the results of an experimental investigation of synchronization of noise-amplifying dynamical systems consisting of nonlinear electronic periodic oscillators. In our experiments, the dynamical behavior of the coupled oscillators is described by a set of nonlinear differential equations given by

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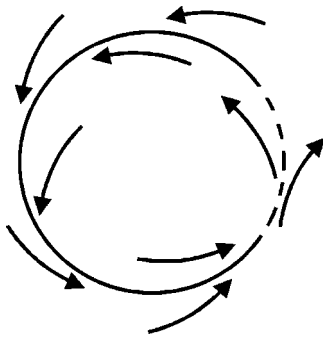


FIG. 1. Schematic diagram of a globally stable limit cycle with varying local stability. The segment of the limit cycle that is locally unstable (dotted line) allows perturbations to grow transiently before they decay asymptotically in the locally stable segment (solid line).

$$\dot{\mathbf{x}}_m = \mathbf{F}[\mathbf{x}_m], \quad (1a)$$

$$\dot{\mathbf{x}}_s = \mathbf{F}[\mathbf{x}_s] + \gamma \mathbf{K}(\mathbf{x}_m - \mathbf{x}_s), \quad (1b)$$

where $\mathbf{x}_m(\mathbf{x}_s)$ denotes the position in the N dimensional phase space of the master (slave) oscillator, $\mathbf{F}$ is a vector field describing the dynamics of an individual oscillator, $\mathbf{K}$ is an $N \times N$ coupling matrix, and γ is the scalar coupling strength. Note that the coupling is one-way so that the evolution of the master oscillator is unaffected by that of the slave. The synchronized state $\mathbf{x}_s(t) = \mathbf{x}_m(t)$ defines an N dimensional invariant manifold (called the *synchronization manifold*) residing in the $2N$ dimensional phase space of the coupled system. We introduce new coordinates $\mathbf{x}_{\parallel} = (\mathbf{x}_m + \mathbf{x}_s)/2$ and $\mathbf{x}_{\perp} = (\mathbf{x}_m - \mathbf{x}_s)/2$ to separate the dynamics along and transverse to the synchronization manifold. In this coordinate system, synchronization may be defined as $\mathbf{x}_{\perp}(t) = 0$. In each of two experiments, we couple two periodic oscillators subject to small input noise and observe the degree of synchronization as a function of coupling strength. We use various norms of $\mathbf{x}_{\perp}(t)$ to quantify the degree of synchronization between the oscillators.

In the first experiment, each oscillator is a one-dimensional (1D) periodically driven system. The phase space trajectory of a single oscillator brings the system very close to a threshold beyond which the trajectory changes shape considerably. A small perturbation that pushes the oscillator over the threshold is amplified as the oscillator follows the new trajectory. When two such oscillators are coupled and the coupling strength is increased from zero, we find that high-quality synchronization occurs suddenly once a critical coupling is reached.

In the second experiment, the oscillators evolve on a limit cycle along which the Jacobian is non-normal at all points. The non-normality gives rise to transient amplification of perturbations as described previously. When two such oscillators are coupled and the coupling strength is increased from zero, we do not observe any threshold for synchronization. The degree of synchronization increases smoothly but slowly. We do not observe high-quality synchronization at any coupling strength in the range of experimentally accessible coupling strengths.

A second objective of this paper is to explore the possible implications of our experiments with noise-amplifying

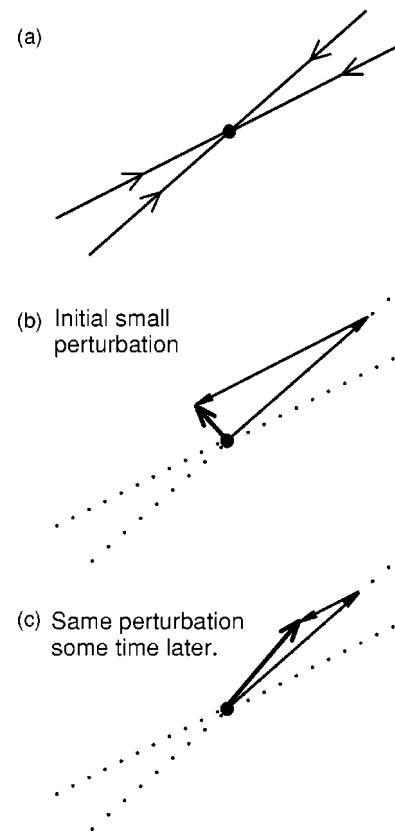


FIG. 2. (a) Schematic diagram of a fixed point in a non-normal linear system. The system has negative eigenvalues and non-orthogonal eigenvectors. (b) A perturbation from the fixed point depicted (in bold) as a superposition of the eigenvectors. The perturbation has a small norm due to cancellation of the nonorthogonal eigenvectors. The dotted lines show the eigendirections of the fixed point. (c) The same perturbation some time later. Both eigenvectors have contracted but at different rates. Cancellation is lost and the norm of the perturbation grows.

systems on recent experimental studies of synchronized chaotic oscillators. Several researchers have reported experiments where, in a regime where synchronization was expected, intervals of synchronization are interrupted irregularly by large, brief desynchronization events (or bursts).³⁻⁶ This bursting is attributed to the interaction of very small noise or parameter mismatch between oscillators and local variations in the stability of the synchronized state. Specifically, the synchronized state may be locally unstable near unstable periodic orbits embedded in the chaotic attractor on which the oscillators evolve. A perturbation to the synchronized motion in the neighborhood of such an orbit may be amplified before decaying asymptotically.³ With this scenario in mind, several researchers have attempted to develop criteria for “high-quality,” “robust,” or “burst-free” synchronization.³⁻⁷ Since none of these criteria explicitly require the oscillators to be chaotic for applicability, we apply them to our experimental systems and compare their predictions with our observations. We find that none of these criteria accurately predicts the behavior observed in our experiments.

In the next two sections, we describe our experimental systems. Section II describes a pair of coupled driven oscillators. We find that these oscillators display high-quality syn-

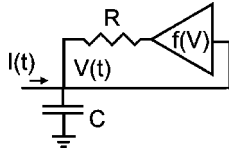


FIG. 3. Schematic diagram of an individual driven nonlinear oscillator. The device labeled $f(V)$ is a high input impedance nonlinear amplifier.

chronization for coupling strengths above a critical value. Section III describes a pair of coupled limit cycle oscillators characterized by highly non-normal Jacobians. These oscillators do not display high-quality synchronization for any observed coupling strength. In Sec. III, we review several proposed criteria for synchronization and apply them to our system. Finally, we discuss the implications of our experiments in Sec. IV.

II. EXPERIMENT 1: 1D DRIVEN OSCILLATORS

The choice of oscillators used in our first experiment is motivated by the noise-induced bursting observed in coupled “double scroll” chaotic oscillators in a previous study.⁵ The “double scroll” chaotic attractor has a saddle point at the origin separating the two “scrolls.” Consider the case when two such oscillators are coupled as master and slave. When synchronized, the master and slave follow identical phase space trajectories. However, a perturbation occurring when the oscillators are synchronously evolving in the neighborhood of the saddle may kick the slave over onto the other scroll for a long excursion away from the master. We mimic this behavior with two well-matched, one-way coupled nonlinear RC circuits driven by identical sinusoidal signals. Note that these oscillators are not chaotic but display a mechanism for noise amplification similar to that found in the double scroll chaotic system. A single oscillator is shown schematically in Fig. 3 and its behavior is described by

$$\frac{dV}{dt} = [f(V) - V]/RC + I(t)/C, \quad (2)$$

where

$$f(V) = \begin{cases} -V_0, & V < -V_0/G \\ GV, & -V_0/G \leq V \leq V_0/G \\ V_0, & V > V_0/G \end{cases} \quad (3)$$

and $I(t) = I_0 + I_d \cos(\omega_d t)$ is a sinusoidal driving current. The parameter values are $R = 2 \text{ k}\Omega$, $C = 0.047 \text{ }\mu\text{F}$, $G = 3.1$, $V_0 = 0.7 \text{ V}$, $I_d = 449 \text{ }\mu\text{A}$, and $\omega_d/2\pi = 100 \text{ Hz}$.

The dynamical behavior of a single oscillator in the absence of noise is shown in Fig. 4(a) for $I_0 = -186 \text{ }\mu\text{A}$ (solid line), generated by a numerical simulation. A brief, sufficiently large perturbation to the system when the trajectory is in the vicinity of $V = -V_0/G$ (dashed line) causes it to undergo a large excursion away from the orbit before returning, as shown by the dotted line in Fig. 4(a). Once the trajectory crosses the threshold, the growth rate of the perturbation is very large: $V(t)$ increases from -0.23 V to $\sim 1 \text{ V}$ in $\sim 0.1 \text{ ms}$ while the period of the driving signal is 10 ms . Thus, a perturbation during the brief interval when the trajectory is in

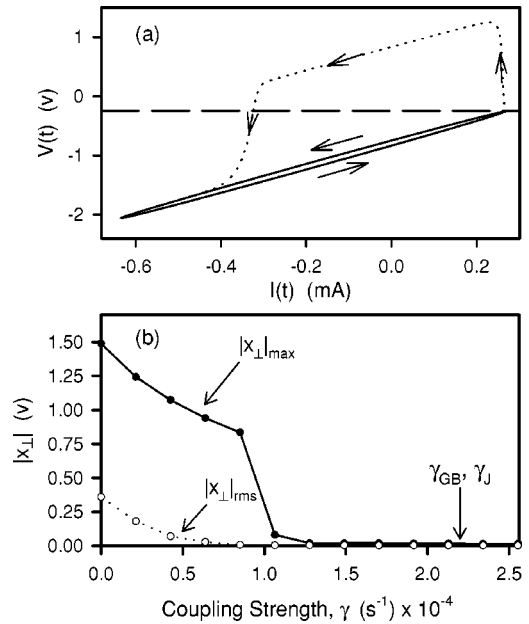


FIG. 4. Synchronization of the driven nonlinear electronic circuit. (a) Numerical simulation of the oscillator trajectory for $I_0 = -186 \text{ }\mu\text{A}$ under noise free conditions (solid line). A perturbation to the system that pushes it across the threshold (dashed line) causes it to undergo a large excursion (dotted line). (b) Two measures of the degree of synchronization as a function of the coupling strength.

the neighborhood of the threshold can be amplified significantly. This behavior resembles the bursting observed in coupled chaotic double scroll oscillators evolving near the saddle point at the origin of their phase space, as discussed previously.

In the experiment, the slave oscillator is coupled to the master by injecting a current $I_{\text{sync}} = \gamma C(V_m - V_s)$ into the slave circuit at the same node as the drive signal. We bias both oscillators very close to the threshold by setting $I_0 = -186 \text{ }\mu\text{A}$. For this value of I_0 and the inherent level of noise in the system, the master oscillator never crosses the threshold and remains on the trajectory shown as the solid line in Fig. 4(a). A small Gaussian white noise current (bandwidth from 10 Hz to 1 kHz , rms current $\sim 0.5\%$ of I_d) is injected into the slave oscillator. When there is no coupling ($\gamma = 0$), the slave occasionally crosses the threshold and bursts away from the periodic orbit. For the oscillators to be synchronized, the coupling has to be chosen so that the slave never undergoes a burst.

For each of several different values of the coupling strength, we record a long time series of the Euclidean norm $|x_\perp| = |V_m - V_s|$. To quantify the degree of synchronization, we determine from these time series the average distance from the synchronization manifold $|x_\perp|_{\text{rms}}$ and the maximum observed value of the distance from the manifold $|x_\perp|_{\text{max}}$ (Ref. 5) for each coupling strength, as shown in Fig. 4(b). For coupling strengths between 0.6 and $0.8 \times 10^4 \text{ s}^{-1}$, $|x_\perp|_{\text{max}}$ is on the order of the size of the orbit ($\sim 2 \text{ V}$) even though $|x_\perp|_{\text{rms}}$ is very small ($\sim 1\%$ of the orbit size), implying that there exist large, brief, occasional desynchronization events even when the oscillators are synchronized on average. From the figure, it is seen that the large desynchroniza-

tion events only cease for $\gamma \geq 1.3 \times 10^4 \text{ s}^{-1}$, as indicated by the large drop in $|x_{\perp}|_{\max}$.

To interpret these results, we must keep in mind that there is no precise way to define synchronization in an experiment. Gauthier and Bienfang referred to synchronization as high quality when $|x_{\perp}|_{\max} < \epsilon$ where ϵ is a small length scale (typically within a few percent of the characteristic dimensions of the attractor).⁵ This condition inherently depends on the choice of a metric and can be violated simply by making the noise level large enough even when the coupled system does not amplify perturbations. In light of this trivial case, this condition should not be taken as a formal definition but rather as an attempt to quantify the idea that small noise or mismatch in the systems should only give rise to small deviations between the master and slave. Without making an explicit choice for ϵ , we see from Fig. 4(b) that the behavior makes a transition from poor synchronization, where small noise gives rise to large separations between master and slave, to burst free synchronization around $\gamma \approx 1.3 \times 10^4 \text{ s}^{-1}$ where small noise causes only small separations. Note that the coupling strength at which this transition occurs increases with increasing level of noise injected into the slave oscillator.

III. EXPERIMENT 2: NON-NORMAL LIMIT CYCLE OSCILLATORS

The choice of oscillators used in our second experiment is motivated by the increasing evidence that noise amplification due to non-normality plays an important role in many chaotic physical systems. For example, non-normality may be responsible for turbulence in some fluid flows⁸⁻¹⁰ and chaotic behavior in mode-locked lasers.¹¹ The non-normality associated with homoclinic tangencies in nonhyperbolic chaotic attractors gives rise to noise-induced attractor deformation,^{12,13} a phenomenon recently observed experimentally in an electronic circuit.¹⁴ Almost all chaotic systems of physical interest are nonhyperbolic and therefore may have regions of phase space characterized by non-normality.¹³ The periodic oscillator used in our second experiment evolves on a stable limit cycle on which the Jacobian is highly non-normal. The experimental apparatus consists of two one-way coupled, well-matched electronic oscillators. An individual oscillator, shown schematically in Fig. 5, is described by the set of equations

$$\begin{aligned} \frac{dx}{dt} &= [-y + g(x - z \cot \alpha) + \lambda z \cot \alpha] / RC, \\ \frac{dy}{dt} &= (x - z \cot \alpha) / RC, \\ \frac{dz}{dt} &= \frac{\lambda}{RC} z, \end{aligned} \quad (4)$$

where

$$g(u) = \begin{cases} -(2+u), & u < -1 \\ u, & -1 \leq u \leq 1 \\ (2-u), & u > 1 \end{cases} \quad (5)$$

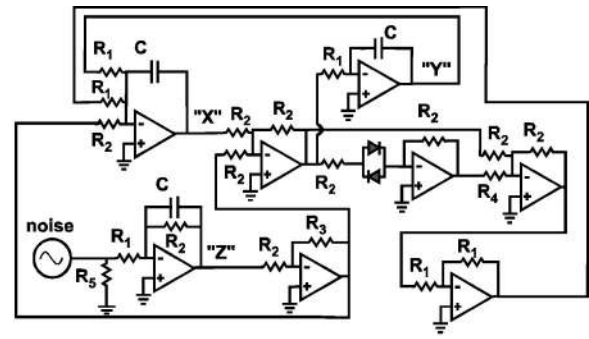


FIG. 5. Schematic diagram of an individual oscillator. Component values are $R_1 = 100 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $R_3 = 150 \text{ k}\Omega$, $R_4 = 5 \text{ k}\Omega$, $R_5 = 50 \Omega$, $C = 1 \text{ nF}$. The diodes are type MV2101. The op amps are type OP-07.

and $\lambda = -10$, and $RC = 0.1 \text{ ms}$. The trajectory of a single oscillator is a limit cycle of radius $\sim 4 \text{ V}$ in the xy plane centered on the origin.

At each point on the limit cycle, the oscillator's Jacobian has two eigenvectors in the xy plane and a third inclined at an angle α with respect to the plane. The eigenvalue associated with this third eigenvector is negative and equal to λ . The nonorthogonality of the eigenvectors (when $\alpha \neq 90^\circ$) indicates the Jacobian is non-normal, giving rise to the possibility that perturbations can undergo significant transient growth. Therefore, perturbing the oscillator in the z direction results in a transient excursion from the limit cycle, the size of which depends on the size of the perturbation and the degree of non-normality. Thus, very small noise can be amplified dramatically by the system with strong non-normality.

We note, as an aside, that a change of coordinates can remove the non-normality from the system, but it cannot remove the noise sensitivity. For example, consider adding a noise term to the evolution equation for the z variable, as done in our experiment, and make the change of variables $z' = z \cot \alpha$ in order to reduce the non-normality. In this new coordinate system, the noise term is multiplied by a factor $\cot \alpha$, which is large in the case of strong non-normality. Thus, the sensitivity to noise is independent of the chosen coordinate system.

In the experiment, the two oscillators are "xy" coupled ($K_{11} = K_{22} = 1$, $K_{ij} = 0$ otherwise). As before, we quantify the degree of synchronization by measuring $|x_{\perp}|_{\max}$ and $|x_{\perp}|_{\text{rms}}$ for several coupling strengths as shown in Fig. 6. To demonstrate the role of non-normality in amplifying noise, the experiment is performed once with very small ($\cot \alpha = 1$) and once with rather large ($\cot \alpha = 100$) non-normality. Gaussian white noise (30 mV rms, dc to 15 MHz) is added to the z component of the slave circuit. The amount of noise added to the slave oscillator is the same in both situations.

For small non-normality ($\cot \alpha = 1$), it is seen in Fig. 6(a) that the observed distance from the synchronization manifold $|x_{\perp}|_{\max} \leq 0.1 \text{ V}$, or 2.5% of the limit cycle radius, for $\gamma > 0.2 \times 10^3 \text{ s}^{-1}$. Apparently, the perturbations are not amplified significantly when $\cot \alpha = 1$ since the eigenvectors are almost orthogonal. We find dramatically different results when the non-normality is increased ($\cot \alpha = 100$), as seen in Fig. 6(b). Over the range of experimentally attainable cou-

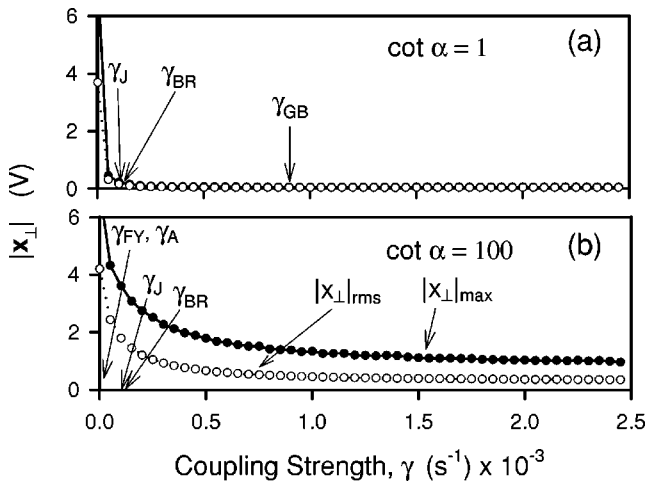


FIG. 6. Synchronization of the “non-normal limit cycle” electronic circuit. Two measures of the degree of synchronization for (a) small ($\cot \alpha = 1$) and (b) large ($\cot \alpha = 100$) non-normality.

pling strengths, $|x_{\perp}|_{\text{max}} \geq 0.9$ V, or 23% of the radius of the limit cycle, and $|x_{\perp}|_{\text{rms}} \geq 0.35$ V, or 9% of the radius of the limit cycle. Thus a high degree of synchronization is never observed. At all accessible coupling strengths, the noise is amplified dramatically. In addition, the degree of synchronization increases slowly and smoothly with no sharp transition as observed in the previous experiment.

IV. CRITERIA FOR SYNCHRONIZATION

In the experiments described above, we investigate the effects of noise on the synchronization of periodic dynamical systems. These oscillators are designed, however, to mimic effects previously observed or theorized in chaotic systems. Specifically, the mechanisms for noise amplification built into these oscillators can also be found in chaotic dynamical systems of physical interest. To further explore the implications of noise-amplifying dynamics for chaotic systems, we apply several criteria for synchronization developed with chaotic systems in mind. Note that none of these criteria requires chaos for applicability. In the next few paragraphs we briefly review each of the five criteria. Then we apply the criteria to our experiments and compare their predictions to the experimental observation reported in Secs. II and III.

Criterion (1). Fujisaka and Yamada¹⁵ claim that the synchronized state of noise free, identical coupled chaotic oscillators is asymptotically stable when the largest transverse Lyapunov exponent is negative. The exponents are determined from the solution of the equations

$$\frac{d\delta\mathbf{x}_{\perp}}{dt} = \mathbf{J}\delta\mathbf{x}_{\perp}, \quad (6)$$

$$\mathbf{J} \equiv D\mathbf{F}[\mathbf{x}_m(t)] - \gamma\mathbf{K}, \quad (7)$$

obtained by linearizing Eq. (1) about $\mathbf{x}_{\perp} = 0$, where $\delta\mathbf{x}_{\perp}$ is a small perturbation away from the synchronization manifold and $D\mathbf{F}[\mathbf{x}_m(t)]$ is the Jacobian of the vector field $\mathbf{F}$ evaluated on the attractor of the master oscillator. Apparently, Fujisaka and Yamada believed this criterion is also valid for systems subject to small noise or parameter mismatch: They incorpo-

rated this criterion into a proposed method for determining experimentally the largest Lyapunov exponent of a dynamical system. This method relies on the coincidence of the coupling strength at which synchronization becomes unstable and the coupling strength at which synchronization is lost experimentally. Many other researchers have assumed that the asymptotic stability of the synchronization manifold is a good predictor of synchronization in real physical systems despite the presence of noise and parameter mismatch. For example, a demonstration that the largest Lyapunov exponent is negative is presented as proof that synchronization of chaotic oscillators will occur (see, e.g., Refs. 16 and 17). Note that several researchers have already shown this criterion to be unreliable.^{3–6} We include it here for completeness.

Criterion (2). Ashwin, Buescu, and Stewart³ suggest that bursts of desynchronization events in coupled chaotic oscillators are due to transversely unstable invariant sets embedded in the transversely stable chaotic attractor on the synchronization manifold. When the system is in the neighborhood of such a set, small noise can push it off the manifold resulting in a brief desynchronization event. Ashwin, Buescu, and Stewart named this behavior *attractor bubbling*. To avoid attractor bubbling, the largest transverse Lyapunov exponent characterizing the most unstable invariant set must be negative for synchronization in the presence of noise.

Criterion (3). Based on the same idea that the stability of unstable sets governs the region of high-quality synchronization, Brown and Rulkov¹⁸ suggest an alternative method for determining the transverse stability of these sets. They developed a sufficient, but not necessary, condition for the asymptotic stability of the synchronized state using Gronwall’s theorem. Briefly, they decompose the matrix $\mathbf{J}$ into a time-independent $\mathbf{A} \equiv \langle D\mathbf{F} \rangle - \gamma\mathbf{K}$ and a time-dependent $\mathbf{B}(\mathbf{x}_m; t) \equiv D\mathbf{F}[\mathbf{x}_m(t)] - \langle D\mathbf{F} \rangle$ parts, where $\langle \cdot \rangle$ denotes a time average over the driving trajectory. A trajectory is transversely stable when

$$-\text{Re}[\Lambda_1] > \langle \|\mathbf{P}^{-1}[\mathbf{B}(\mathbf{x}_m; t)]\mathbf{P}\| \rangle, \quad (8)$$

where Λ_1 is the largest eigenvalue of $\mathbf{A}$ and $\mathbf{P}$ is a matrix of eigenvectors of $\mathbf{A}$. The notation $\|\cdot\|$ denotes a norm whose choice is arbitrary. The predictions of this criterion depend both on the choice of norm and metric. Following Brown and Rulkov, we use the Frobenius norm. As for the previous criterion, Brown and Rulkov suggest evaluating this criterion along every invariant set embedded in the attractor in order to determine the region of high-quality synchronization.

Criterion (4). Pecora, Carroll, and Heagy¹⁹ and Johnson et al.²⁰ attempt to ensure high-quality synchronization by requiring all eigenvalues of the matrix $\mathbf{J}$ have negative real parts at all points along the driving trajectory $\mathbf{x}_m(t)$. As for the previous case, this criterion depends on the choice of metric.

Criterion (5). Gauthier and Bienfang⁵ introduce the Lyapunov function $\mathcal{L} \equiv |\delta\mathbf{x}_{\perp}(t)|^2$ to obtain an estimate of the regime of high-quality synchronization. They suggest that high-quality synchronization occurs for coupling schemes where

$$\frac{d\mathcal{L}}{dt} = 2 \delta \mathbf{x}_\perp(t) \cdot \mathbf{J} \delta \mathbf{x}_\perp(t) < 0 \quad (9)$$

for all times. An equivalent statement is that all eigenvalues of the matrix $(\mathbf{J} + \mathbf{J}^T)$ have negative real parts at all points along the driving trajectory. In effect, this criterion requires that all perturbations transverse to the synchronization manifold must decay to the manifold without transient growth.²¹ As in the two previous cases, this criterion depends on the choice of metric.

Applying the criteria to our first experiment, we find that each predicts synchronization for all coupling strengths greater than a critical value. Although we do see a sharp transition to high-quality synchronization in the experiment, none of the criteria accurately predict the coupling strength $\gamma \approx 1.3 \times 10^4 \text{ s}^{-1}$ at which this transition is observed. We denote the critical coupling strength for criteria 1–5 as γ_{FY} , γ_{A} , γ_{BR} , γ_{J} , and γ_{GB} , respectively. Our analysis of the circuit model reveals that $\gamma_{\text{FY}} = -1.110^4 \text{ s}^{-1}$. We attribute the negativity of γ_{FY} to the fact that, without the driving currents, the nonlinear RC circuits do not self-oscillate. Since the master and slave receive identical driving currents, the circuits would behave identically even with zero coupling if no noise were present. Criterion (2) is equivalent to criterion (1) in this case since the only invariant set in the attractor is the orbit; thus, $\gamma_{\text{A}} = -1.1 \times 10^4 \text{ s}^{-1}$. In addition, for the criterion proposed by Brown and Rulkov (3), we find $\gamma_{\text{BR}} = -1.1 \times 10^4 \text{ s}^{-1}$. From Fig. 4(b), it is clear that these three criteria fail to provide even a “sufficient” condition for high-quality synchronization since $|x_\perp|_{\text{max}}$ and $|x_\perp|_{\text{rms}}$ are large for a range of positive coupling strengths below $\gamma \approx 1.3 \times 10^4 \text{ s}^{-1}$. On the other hand, criteria (4) and (5) predict $\gamma_{\text{J}} = \gamma_{\text{GB}} = 2.2 \times 10^4 \text{ s}^{-1}$, overestimating the coupling strength needed to obtain high quality synchronization in the experiment for the particular level of noise injected into the slave oscillator.

Applied to our second experiment, each criterion again predicts a critical coupling strength above which high-quality synchronization should be observed. We first consider the case of small non-normality where experimentally we observe high-quality synchronization at coupling strengths greater than $0.2 \times 10^3 \text{ s}^{-1}$. For the first four criteria, we find that $\gamma_{\text{FY}} = \gamma_{\text{A}} = 0$, $\gamma_{\text{BR}} = 0.13 \times 10^3 \text{ s}^{-1}$, and $\gamma_{\text{J}} = 0.11 \times 10^3 \text{ s}^{-1}$. The fifth criterion predicts a much larger critical coupling strength of $\gamma_{\text{GB}} = 0.9 \times 10^3 \text{ s}^{-1}$. Thus, the first four criteria reasonably predict the range of coupling strengths over which a high degree of synchronization is observed while the fifth criterion significantly overestimates the required coupling strength. More interesting is the case of large non-normality where we observe experimentally no transition to high-quality synchronization. Applying the criteria in this case, we find that the first four criteria are independent of the non-normality and hence predict high-quality synchronization at the same critical coupling strengths as in the previous case despite the fact that the observed degree of synchronization is degraded significantly. On the other hand, we find the fifth criterion predicts $\gamma_{\text{GB}} = 6.9 \times 10^6 \text{ s}^{-1}$, a value much too large to implement using our experimental apparatus. We attribute the sensitivity to non-normality in the sys-

tem of criterion (5) to the above-mentioned fact that this criterion restricts transient amplification of perturbations.

V. DISCUSSION

From these experiments we draw two conclusions. First, any physical system that contains some mechanism for transient growth of perturbations may show bursting behavior when coupled to an identical system because of the inevitable presence of noise. In this paper, we have presented two such mechanisms for transient growth: (1) a sharp threshold in phase space that can separate the master and slave, and (2) noise amplification due to non-normality. Note that these are not the only possible mechanisms for transient growth. Other examples are attractor bubbling in coupled chaotic oscillators³ and local stability variations in limit cycles.¹ Therefore, a general criterion for synchronization of physical systems must have some sensitivity to transient behavior. In our experiments, only criterion (5) shows any sensitivity to the transient growth displayed by the oscillators, but it appears to be overly conservative in its estimate of the coupling strength needed for high-quality synchronization. Second, the details of the bursts, their size and frequency, may depend intimately on the details of the noise. For example, in our first experiment, the frequency of the bursts is directly determined by the frequency of perturbations that are both large enough and time appropriately. We speculate that further progress in predicting high-quality synchronization will require explicitly taking into account the details of the noise in the system.

ACKNOWLEDGMENTS

J.N.B. and D.J.G. gratefully acknowledge the financial support of the U.S. ARO through Grant Nos. DAAD 19-99-1-0199 and DAAG55-97-1-0308.

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